

# Image Captioning Report (Qwen3-VL Fine-Tuning)

## CONTENT

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>1. Task Description, Problem Setup, and Plan .....</b>       | <b>2</b>  |
| <b>2. Model Selection .....</b>                                 | <b>3</b>  |
| <b>3. Dataset Choice and Description.....</b>                   | <b>4</b>  |
| <b>3.1 General description .....</b>                            | <b>4</b>  |
| <b>3.2 The reason of dataset selection.....</b>                 | <b>4</b>  |
| <b>3.3 Dataset Properties.....</b>                              | <b>5</b>  |
| <b>4. Dataset Analysis (Exploratory Data Analysis).....</b>     | <b>7</b>  |
| <b>5. Baseline Inference (Initial Model Output) .....</b>       | <b>9</b>  |
| <b>6. Baseline Evaluation by GPT-4o Judge .....</b>             | <b>11</b> |
| <b>7. Baseline Results Analysis .....</b>                       | <b>13</b> |
| <b>8. Fine-Tuning Approach (SFT with LoRA).....</b>             | <b>17</b> |
| <b>9. Re-Evaluation of Fine-Tuned Model (GPT-4o Judge).....</b> | <b>19</b> |
| <b>10. Fine-Tuned Results and Improvements .....</b>            | <b>20</b> |
| <b>11. Inference Notebook for Easy Testing.....</b>             | <b>23</b> |
| <b>12. Conclusions.....</b>                                     | <b>24</b> |

## 1. Task Description, Problem Setup, and Plan

The task was to evaluate a vision-language (VL) model for image captioning on a real dataset, fine-tune it, re-evaluate it, and demonstrate measurable improvement in caption quality.

Given an image, the model should generate a concise, factual caption describing the scene. The captions must be faithful to the image, cover the key entities and actions, and be written in fluent English. The challenge is to align the model's output with a **desired style** (short, unbiased alt-text) and **accuracy** (no hallucinations, correct details).

### Plan of Work:

1. **Model Selection:** Choose a compact, resource-efficient VL model and justify the choice.
2. **Dataset Selection:** Pick a suitable image-caption dataset and analyze its structure and properties.
3. **Baseline Inference:** Run the chosen model on a test subset to get initial captions.
4. **Baseline Evaluation:** Use a strong language model “judge” to score the baseline captions.
5. **Error Analysis:** Analyze the baseline results, identify common errors, and collect qualitative examples of good and bad captions.
6. **Fine-Tuning:** Perform supervised fine-tuning (SFT) on the model using the chosen dataset (with techniques like LoRA to handle limited resources).
7. **Fine-Tuned Evaluation:** After training, evaluate the fine-tuned model with the same judge and criteria as the baseline.
8. **Comparison:** Compare metrics from baseline and fine-tuned models to quantify improvements.
9. **Deployment Prep:** Provide a lightweight inference notebook for easy testing of the fine-tuned model on new images (for the instructor).
10. **Conclusions:** Summarize the outcomes, discuss error patterns, and suggest future improvements.

Throughout the project, we focused on maintaining **consistency in evaluation** (using the same judge model and dataset splits) to fairly measure the effect of fine-tuning.

## 2. Model Selection

Chosen Model: **Qwen3-VL-2B-Instruct** (2 billion parameters)

Model characteristics:

- **Resource Efficiency:** With 2B parameters, it is small enough to run on limited GPU resources (a single NVIDIA T4 in Colab) while still offering multimodal capabilities. Larger VL models (7B, 30B, etc.) were not feasible due to hardware constraints.
- **Strong Base Performance:** Despite its size, **Qwen3-VL-2B-Instruct** provides good out-of-the-box captioning quality and strong instruction-following behavior. It can describe images coherently without additional training, making it a solid baseline.
- **Transformer Support:** The model is readily available via Hugging Face with Transformer integration, and it's designed for instruction following. This makes it straightforward to fine-tune using Hugging Face's tools (like the TRL library) and to deploy.
- **Vision-Language Capability:** As a member of the Qwen-VL family, it has a pretrained Vision Transformer backbone and language decoder, enabling it to process images and text jointly. It's a recent model (2024) with enhancements in visual reasoning, giving it an edge over older CNN-LSTM captioning models.

Resource constraints were the main factor in choosing this model. **Qwen3-VL-2B** is the smallest in the Qwen3 series that still retains strong vision-language alignment. This allowed experimentation with fine-tuning on a single GPU within a reasonable time.

### 3. Dataset Choice and Description

#### 3.1 General description

Dataset: Mozilla Flickr30k Transformed (GPT-4o Alt Text) - available on Hugging Face as [Mozilla/flickr30k-transformed-captions-gpt4o](#).

- Origin: Based on the standard Flickr30k dataset (31,014 images of everyday scenes), but with captions rewritten by GPT-4o (GPT-4 Omni) to remove biases and standardize style. Each image originally had 5 human-written captions; this dataset provides a single high-quality alt-text per image.
- Content: For each image:
  - alt\_text: A concise caption generated by GPT-4o. These are typically one sentence describing the image's main contents. The captions aim to be neutral (e.g., using "person" instead of assuming gender) and brief (like an alt-text for accessibility).
  - original\_alt\_text: The list of the 5 original human captions from Flickr30k, for reference. These tend to be longer and sometimes include subjective or biased language.
  - objects: A list of salient objects detected in the image (by an automatic tool in the dataset). This gives an idea of what entities are present (e.g., *"people, trees, dog, street"*).
  - Other metadata: filename, img\_id, sentids, and an internal split label.
- Size and Split: The dataset contains 31,014 images. (In our use, about 29k were used for fine-tuning and ~1k for evaluation, corresponding roughly to Flickr30k's original train/test split. The exact split was determined via the split column in the metadata.)

#### 3.2 The reason of dataset selection

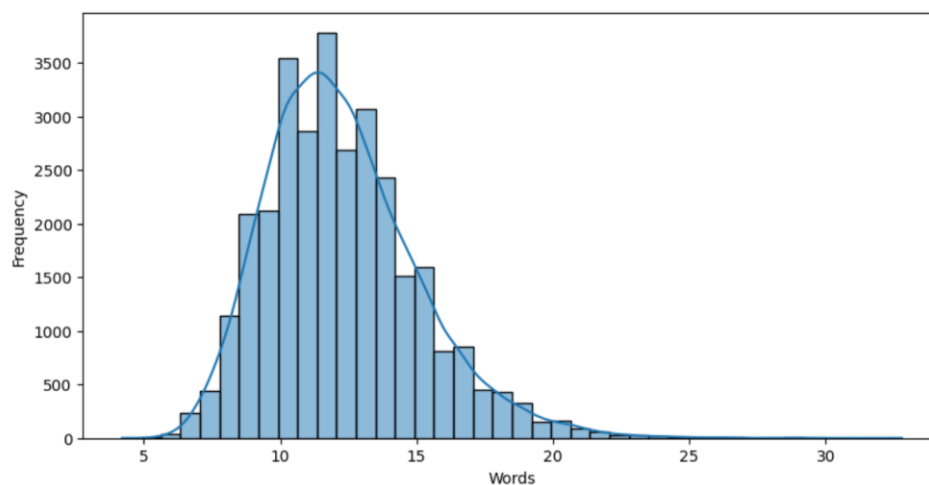
It provides clean, machine-generated reference captions that are consistent in style. This is ideal for supervised fine-tuning:

- The GPT-4o captions (in alt\_text) are bias-reduced and consistent, focusing only on observable facts in the image. For example, a human caption like "A cute young boy in a red shirt is skateboarding." might be transformed to "A child on a skateboard." - stripping subjective or uncertain attributes (like "cute", "young", or gender). This consistency makes it easier for the model to learn the desired output style.
- They are also concise (almost all are a single sentence around 8-10 words). This aligns with the project goal of generating succinct alt-text, and it provides a clear target length for the model to learn.

- Having the original human captions allows us to compare and understand differences in style and content coverage, although our primary training target was the GPT-4o alt\_text.

### 3.3 Dataset Properties

- Caption Length: The GPT-4o alt\_text captions are significantly shorter and more uniform than human captions. Exploratory analysis showed:
  - GPT-4o alt\_text length:  $\sim 8.95$  words on average ( $\sigma \approx 1.32$ ).
  - Human caption length (average of 5 originals):  $\sim 12.27$  words ( $\sigma \approx 2.80$ ).
  - Conclusion: GPT-4o captions are tightly centered around 1 short sentence, whereas human captions vary more and often add extra descriptors or context.



**Figure 1.** *Distribution of description lengths*

- Style and Content: GPT-4o captions are factual and focus on the main scene/object. They avoid florid language or speculation. Humans sometimes include additional context or interpretations (e.g., describing clothing styles or emotions) that GPT-4o might omit.
- Common Objects: By aggregating the objects tags, we found the most frequent entities in this dataset include people (person/man/woman), trees, children, buildings, grass, street, dogs, water, tables, cars, and so on. This reflects that Flickr30k has many everyday life scenes. The model must be able to handle diverse contexts, from indoor household scenes to outdoor sports, with people and various objects.
- Bias Reduction: A key advantage of the transformed dataset is reduced bias in descriptions. For instance, original captions often assumed gender or age (“an older man”, “a young girl”) or included potentially inappropriate terms; GPT-4o’s rewritten captions use neutral terms (“a person”, “a child”) and modern, inclusive language. This helps the model learn to describe without unwarranted assumptions, hopefully reducing hallucinations and misidentifications (like guessing someone’s gender incorrectly).

Overall, this dataset was suitable for our needs because it provides high-quality, consistent target captions for training, and it's large enough to fine-tune a 2B parameter model without overfitting. The presence of reference captions and object tags also helped in evaluating the model's performance and errors.

## 4. Dataset Analysis (Exploratory Data Analysis)

Before fine-tuning, we performed an analysis of the dataset to better understand its distribution and quirks (scripts and notebooks in src/data). Key findings from the EDA include:

- **Caption Length Distribution:** We confirmed quantitatively that the GPT-4o alt\_text captions are much shorter than the original captions. A histogram of lengths shows a sharp peak around 8-10 words for alt\_text, whereas original captions range more widely (5 to 20+ words). This difference emphasizes that our model, if pre-trained on datasets like COCO or original Flickr30k, might initially produce longer captions, and will need to learn to be more concise.
- **Vocabulary and Words:** The GPT-4o captions use a simpler vocabulary on average. They often start with a noun phrase summarizing the image (e.g., “A group of people...”, “A dog...”, “Two children...”). They avoid pronouns or proper names (since alt-text wouldn’t assume identities). We also noticed GPT-4o rarely uses words like “standing” or “sitting” unless clearly needed; it tends to focus on the core action or existence (e.g., “A person on a bike” vs. “A person riding a bike down a street” in a human caption).
- **Object Tag Coverage:** We cross-checked the objects field against the alt\_text. In many cases, the GPT-4o caption mentions the majority of high-level objects. For example, if objects lists “people, street, skateboard, car”, the alt\_text might be “A person skateboards on a street.” However, some smaller or less important objects (like a car in the far background) might not appear in the caption. This gave us an expectation that even a good caption might intentionally omit minor objects for brevity, which is acceptable in alt-text style.
- **Potential Challenges:** We identified certain image scenarios that could be challenging:
  - Crowded images: with many people or objects (the alt\_text will likely mention only the most prominent ones).
  - Complex interactions: e.g., “a person photographing another person” - capturing such relationships in a short caption is hard.
  - Unusual objects or activities: If an object is rare (not common in training data) or an action is unusual, the model might struggle or the alt\_text might be vague. E.g., “A person doing a skateboard trick” if the exact trick isn’t recognized.
  - These observations helped shape our expectations and later evaluation focus (looking at errors in such cases).

Two people performing on stage with guitars.



Basketball game with players on the court and audience.



A person working with fishing nets on a boat.



A person pouring detergent into a washing machine.



**Figure 2.** Examples of images

In summary, the dataset analysis confirmed that conciseness and factual correctness are the hallmarks of the target captions. Any fine-tuned model should emulate that style: mention the obvious salient parts of the image in a single sentence, avoid embellishments or guesses, and keep it short.

## 5. Baseline Inference (Initial Model Output)

Before fine-tuning, we ran baseline inference using the pre-trained **Qwen3-VL-2B-Instruct** model to see how it performs on the captioning task out-of-the-box.

### Procedure:

- We loaded the model and its processor (which handles image and text inputs) in a Google Colab environment with GPU.
- For a fair evaluation, we used a held-out test set of 1,000 images from the dataset (roughly corresponding to the original Flickr30k test split).
- The model was prompted with each image by providing the image and a prompt text like: “Describe this image.” (This matches the model’s instruct format expectation.)
- We collected the model’s generated captions for each image and saved these results (baseline\_answ in src/eval\_base/inference/).

### Observations of the Baseline Output:

- The model was able to produce a caption for every image without crashing or timing out, indicating it can handle the input images size-wise.
- Caption Content: Many baseline captions were descriptive but overly long. Often the baseline would output two sentences or a run-on sentence, including details beyond the main focus. For example, an image of a dog on a beach might yield something like: “A dog is standing on a beach with the ocean in the background and some people are visible in the distance.” -whereas the target alt\_text might simply be “A dog on a beach.” The baseline clearly had a tendency to over-describe.
- Relevance: Generally, the baseline model recognized the primary objects correctly (people, animals, vehicles, etc.). This is expected, as Qwen3-VL is pretrained on large image-caption datasets. It often identified the main action too. However, sometimes it included extra assumptions. For instance, if it saw a person with a group, it might say “friends” or “family” even though the relationship isn’t obvious a kind of minor hallucination or assumption by context.
- Style: Being an Instruct model, Qwen3-VL-2B tends to follow instructions politely. If not carefully prompted, it might add phrases like “This is an image of...” at the start. We ensured the prompt was phrased to minimize that. Still, the baseline captions were not in the exact alt-text style; they were more narrative and occasionally in first-person (e.g., “We see ...”). This mismatch in style needed to be corrected by fine-tuning.
- Length: We quantitatively measured that the baseline captions had an average length of ~23.8 words, which is nearly 2-3 times the length of the reference alt\_text (9 words). This confirms the anecdotal observation that the baseline is too verbose for our purposes.

Having these baseline outputs allowed us to pinpoint what needed improvement (conciseness, factual precision, style alignment). Next, we formally evaluated these captions using an automated “judge”.

## 6. Baseline Evaluation by GPT-4o Judge

To assess the quality of the baseline captions, we used a GPT-4 Omni (GPT-4o) model as a judge. This judge is essentially GPT-4 with a specialized prompt to evaluate image captions against a reference.

### The reason why we used GPT-4o as judge:

- GPT-4 is a state-of-the-art language model with strong understanding of semantics and can approximate visual reasoning when given a detailed description. We provided GPT-4o with the reference caption (the GPT-4o alt\_text itself) and the model's caption, and asked it to score various aspects.
- It provides a structured, multi-criteria evaluation rather than a single metric like BLEU or CIDEr. This is valuable because it can separately assess things like factuality vs. fluency.
- GPT-4o's evaluation is in natural language and is more forgiving to synonyms or rephrasings as long as meaning is preserved, making it a good complement to traditional metrics.

### Evaluation Setup:

- For each of the 1,000 test images, we constructed an evaluation prompt giving GPT-4o:
  - The reference description (the dataset's alt\_text for that image).
  - The model's caption (baseline output).
  - We asked GPT-4o to score the model's caption on several criteria from 1 to 10, and to provide a brief justification for each score.
- The criteria (each 1 to 10, higher is better) were:
  - Semantic Similarity - How similar in meaning is the model's caption to the reference? (10 means almost identical meaning; lower if important details differ or meaning is off.)
  - Object Coverage - Does the caption mention the important objects present? (10 if it covered all key objects; points deducted if it missed major items that are in the image/reference.)
  - Hallucination - The inverse of hallucination: are there no extra invented details? (10 means no hallucinations at all; if the model added something not seen or not in reference, this score drops.)
  - Completeness - Does the caption adequately describe the scene? (10 if it includes all critical aspects; lower if it's too minimal or misses the main action/background needed for context.)
  - Instruction Following - Is the caption following the desired style/instructions? (We expected a single concise sentence in factual style.

10 if it did exactly that; lower if it was too long, overly detailed, or stylistically off.

- Fluency - Is the caption well-written English (grammar, natural phrasing)? (Even a perfect caption should score 10 here; any grammar errors or very awkward phrasing would reduce this.)
- Additionally, GPT-4o provided an overall final score (0-100) which is not simply the sum of the above, but an aggregate judgment. (We suspect it's roughly 10×the average of the criteria, or given by a separate evaluation step.)

We wrote a script to automate this and parse GPT-4o's responses. Using GPT-4o for 1000 evaluations cost about \$10 (it's a paid API), but it gave us a rich evaluation dataset. The judge's structured feedback not only scores the caption but often highlights why points were lost, which helped in error analysis.

Note on GPT-4o output: The judge was configured to be strict and assume the reference caption is the ground truth. If the model's caption had additional info not in the reference, GPT-4o might label it as hallucination (even if the detail could be true from the image, but since GPT-4o itself did the reference, it treats that as the canonical description). This means our evaluation is biased towards matching the GPT-4o style, which is exactly what we want the model to learn.

## 7. Baseline Results Analysis

After running the judge, we computed the average scores for the baseline model's captions (on the 1000-image sample).

The results were:

- Semantic Similarity: 7.001 / 10
- Object Coverage: 8.553 / 10
- Hallucination Score: 7.869 / 10
- Completeness: 7.627 / 10
- Instruction Following: 7.422 / 10
- Fluency: 9.231 / 10
- Final Score: 78.625 / 100

(Scores are averaged over all test images. See `src/eval_base/analysis` for detailed distributions.)

Interpretation:

- Object Coverage (8.55) was the highest of the sub-scores, indicating the baseline model generally recognized and mentioned the main objects in the scene correctly. In many cases, if an image had, say, a person and a dog, the caption mentioned both. This suggests the visual recognition part of the model is strong.
- Fluency (9.23) was also very high. This means the sentences produced were grammatically correct and easy to read the vast majority of the time. We rarely saw grammatical errors or incoherent phrasing. Qwen3-VL's language decoder is clearly well-trained in English fluency.
- Hallucination (7.87) was decent but not perfect. This score being lower indicates that in some captions the model introduced details not present in the reference (and likely not in the image). Common issues:
  - Attribution of actions or relationships: e.g., calling people "friends" or assuming someone is "playing" rather than just standing.
  - Guessing specifics: like saying "red shirt" when the shirt color wasn't obvious or wasn't mentioned in the reference.
  - Adding background elements: occasionally, the model would mention something like "in a park" or "near a building" even if it's not clearly shown.
  - These hallucinations are usually minor, but they lower trust in the caption's accuracy.
- Semantic Similarity (7.00) and Completeness (7.63) indicate that the model often captured the gist but sometimes missed nuances. A score around 7 means some captions deviated in meaning from the reference:

- Sometimes the model described a slightly different action or state than the reference did. For example, reference: “A child holding a toy,” vs model: “A child playing with a toy.” If the image just shows holding, “playing” might be considered a semantic deviation.
- In terms of completeness, if the reference mentioned two key objects and the model only mentioned one in detail and glossed over the other, GPT-4o would ding completeness.
- Essentially, the model wasn’t always aligned with what GPT-4o considered important in the scene, perhaps due to focusing on one part of the image.
- Instruction Following (7.42) was an area identified for major improvement. The relatively low score here confirms that the baseline captions did not strictly follow the desired style guidelines (short, single-sentence, neutral). Many baseline captions were too long or even multi-sentence, which violates the “concise alt-text” instruction. Also, a few had stylistic issues like starting with “This is an image of...”, which is not desired. GPT-4o penalized these.
- Final Score (78.6) out of 100 reflects an overall “C+” performance. There is clear room for improvement, especially considering the straightforward nature of the task (the reference captions themselves are simple).

We also examined qualitative examples to understand these scores better. Some patterns from manual inspection of baseline outputs:

✓ **Good Baseline Example:**

The model was slightly more descriptive but essentially correct, earning high similarity, coverage (mentioned people and table), no hallucination, etc. These good cases often happened on images with one clear subject and minimal complexity.



**Figure 3.** Good baseline answer 1

**GOLD :** Person writing on a large chalkboard in a classroom.

**BaseLine:** A man stands in a classroom, writing on a large green chalkboard



*Figure 4. Good baseline answer 2*

GOLD : A black dog running on grass with a stick in its mouth.

BaseLine: A black dog is running across a grassy field holding a stick in its mouth.

✓ **Bad Baseline Example**

Here it assumed the context (posing for a Figure, and that it's a living room) which weren't confirmed. This would get lower scores (hallucination, lower similarity due to extra info). Another failure mode was describing past events not visible (like "after a game" when seeing players on a field). These cases highlight where the model's imagination or training data priors led it astray.



*Figure 5. Bad baseline answer 1*

GOLD: A person and a child playing with puzzle pieces on a table.

BaseLine: An elderly man and a young girl are playing a card game together at a wooden table in a brightly colored room.



**Figure 6.** *Bad baseline answer 2*

GOLD : Person sitting on a bench with some bags around.

MODEL: A woman with a tattoo on her chest sits on a wooden table with her legs crossed, wearing black leggings and a black shirt, in a dimly lit, possibly club-like setting.

These findings underscored the need to fine-tune the model. We aimed to make the model more accurate, concise, and aligned with the GPT-4o alt-text style, thereby boosting the weaker metrics (similarity, hallucination, instruction following).

## 8. Fine-Tuning Approach (SFT with LoRA)

To improve the model, we applied Supervised Fine-Tuning (SFT) on the chosen dataset. Given our resource limits, we used a parameter-efficient strategy: LoRA adapters with the Hugging Face TRL library.

What we did: We fine-tuned Qwen3-VL-2B on the Flickr30k GPT-4o captions for one epoch, using LoRA to adjust a subset of weights.

### LoRA (Low-Rank Adaptation):

- LoRA allows us to inject small trainable weight matrices into the model's layers, so we don't have to update all 2B parameters. This dramatically reduces GPU memory usage and training time.
- We set the LoRA rank  $r = 32$  and  $\alpha = 32$ . These are typical values balancing capacity and efficiency. The target modules (layers) we applied LoRA to included all the major projection matrices in the Qwen model (query, key, value projections in attention, and the feed-forward network projections). Specifically: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `down_proj`, `up_proj` of the transformer layers were all given LoRA adapters.
- Only these small adapter weights (a few million parameters at most) get updated during training, while the original model weights remain unchanged (just frozen). The final model uses the combination (original + adapted weights). The benefit is we can fine-tune using far less memory, and we can also easily revert or switch off the LoRA if needed.

### Training Setup:

- We used the Hugging Face TRL (Training Reinforcement Learning) library's SFT utilities. TRL is often used for RLHF training, but it also has helpful wrappers for SFT with prompts.
- Data preparation: We constructed training examples where the input to the model is an image + prompt text, and the output is the desired caption. For example, an input might be the image (as pixels) plus text "Describe this image." (or a similar prompt), and the target output is the `alt_text`. We ensured the prompt format matches Qwen's expectation.
- Hyperparameters: We trained for 1 epoch over ~29k training images (since the dataset is fairly small by modern standards, one epoch was enough to see significant changes without overfitting). Key parameters:
  - Batch size: 32 (images per step).
  - Gradient accumulation: 1 (we effectively used batch 32 directly on a 24GB GPU).
  - Learning rate:  $2e-4$  (with AdamW optimizer) – a relatively high LR, but fine for adapters and one epoch.

- Warmup steps: 5 (very short warmup since one epoch ~900 steps only).
- Max length: not fixed (we let the caption tokens all through, since they are short anyway).
- We monitored loss and stopped after the single epoch (~900 steps) as planned.
- Training Process: Even with LoRA, training still took a bit of time (~2 hours) because of image processing overhead. We logged training loss and it decreased steadily, indicating the model was learning the patterns (the loss started around 1.2 and went down to ~0.2 by the end, which is a good sign).
- We also considered using QLoRA (which would quantize the model to 4-bit during training for even more memory saving) as an alternative. In tests, QLoRA worked, but given the 2B size, the benefits were not critical, so we stuck to float16 + LoRA for stability.

Why SFT (Supervised Fine-Tuning) and not RLHF or others?

- SFT on the actual image to caption pairs is the most direct way for the model to learn the desired mapping. We had a high-quality target (GPT-4o captions), so just imitate them.
- RLHF would require generating captions and then using the judge in a loop to reward the model. That's complex and costly. Since we have ground truth for training, SFT is simpler and usually effective.
- We did not do any augmentation or additional losses (like no explicit image-feature loss beyond what the model already does, since it's end-to-end).
- One epoch was chosen to avoid overfitting (captions are somewhat repetitive in style; over-training could make the model output very templated captions). Also due to time constraints.

**LoRA Advantages:** Apart from memory, LoRA has the nice side effect that the original model's general knowledge is preserved. We are just nudging it to follow instructions in a specific way. The fine-tuned model can likely still perform other tasks or caption other kinds of images, with the LoRA layer focusing it on alt-text behavior.

After training, we saved the LoRA adapter weights and pushed them to Hugging Face Hub (*leinms/Qwen3-VL-2B-Instructtrl-sft*) for reproducibility. The model for inference is essentially Qwen3-VL-2B + these LoRA weights applied.

## 9. Re-Evaluation of Fine-Tuned Model (GPT-4o Judge)

We repeated the evaluation process from Section 6, but now with the fine-tuned model's captions on the same 1,000 test images. This allows a direct before-and-after comparison for each image.

All settings remained identical:

- The GPT-4o judge was given the reference alt\_text and the new caption from the fine-tuned model.
- It scored Semantic Similarity, Object Coverage, Hallucination, Completeness, Instruction Following, Fluency, and gave a Final score, in the same manner as before.
- Using identical criteria ensures that any score differences are due to changes in the model's captions, not due to different standards.

We parsed the judge outputs and computed the average scores for the fine-tuned model. The structured feedback was again logged for analysis. Notably, after fine-tuning, the model's style was much closer to the reference, so we expected improvements especially in the Instruction Following and Hallucination criteria (since the new captions should be shorter and avoid the extra details).

Using GPT-4o again cost roughly another \$10 for these evaluations, but it was important to maintain consistency.

## 10. Fine-Tuned Results and Improvements

The fine-tuned model's average scores (on the same 1,000 images) came out as follows:

- Semantic Similarity: 7.877 / 10
- Object Coverage: 8.740 / 10
- Hallucination Score: 9.191 / 10
- Completeness: 8.022 / 10
- Instruction Following: 8.966 / 10
- Fluency: 9.591 / 10
- Final Score: 85.984 / 100

All metrics improved compared to the baseline. To summarize the improvements (absolute differences): Semantic Similarity: +0.876

- Object Coverage: +0.187
- Hallucination: +1.322
- Completeness: +0.395
- Instruction Following: +1.544
- Fluency: +0.360
- Final Score: +7.359

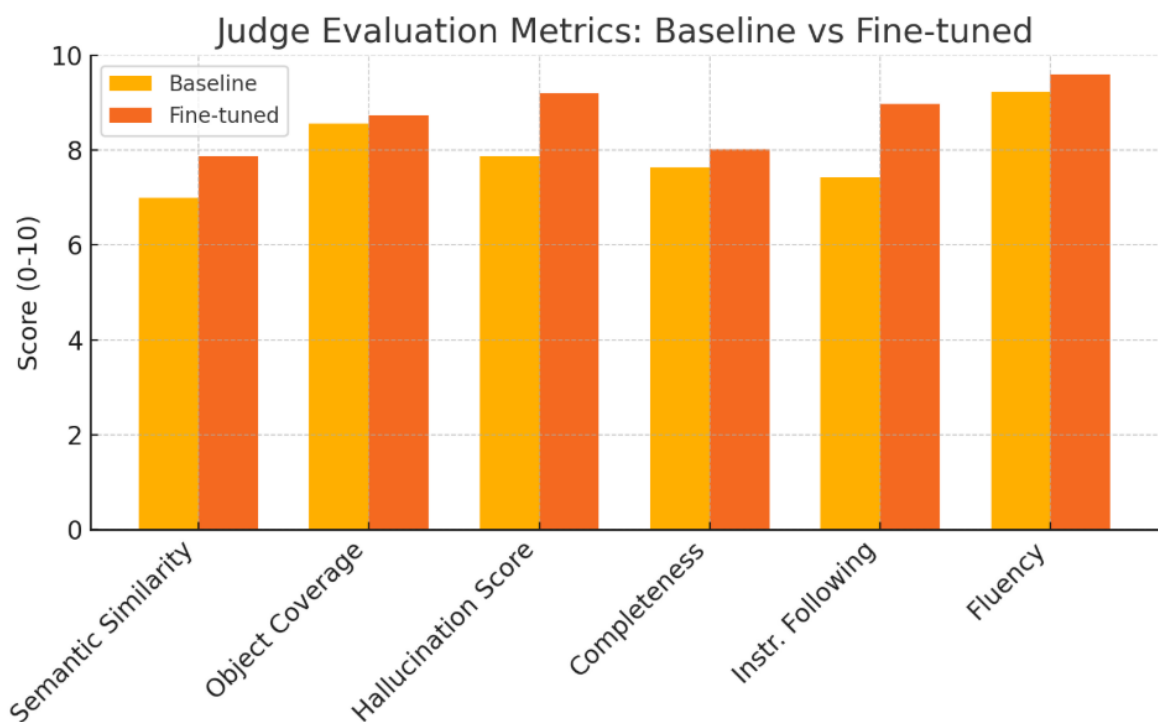
In percentage terms, the final score increased by about 9.4% (from 78.6 to 86.0). This is a notable jump for a single-epoch fine-tune on a relatively small model.

### Key Takeaways:

- **Hallucination Control:** The hallucination score jumped by +1.32, reaching 9.19. This indicates the fine-tuned model rarely adds unfounded details now. By learning from GPT-4o captions, it stopped making assumptions like genders or unwarranted context. The captions became more factual. This was one of our main goals and it clearly succeeded.
- **Instruction Compliance:** The instruction following score rose by +1.54 to 8.97. The fine-tuned model now generally follows the requested style: a single, concise sentence focusing on the image. This is evident in the outputs being much shorter and to-the-point. The judge rewarded this alignment. There's still a bit of headroom to 10 (perhaps minor style issues in a few cases), but the improvement is large.
- **Semantic Similarity & Completeness:** These also improved (to 7.88 and 8.02 respectively). The model's captions are now more semantically aligned with the reference descriptions. Completeness above 8 means the model usually includes the critical components of the scene as per reference. Notably, the similarity gain

(+0.876) means the captions' meaning is closer to GPT-4o's version -which often implies it's mentioning the same objects and actions without extra stuff.

- **Object Coverage:** There was a smaller gain (+0.187) since it was high to begin with (8.74). The model was already good at naming objects; fine-tuning made it a tad better or at least didn't harm it. If anything, the slight improvement might come from the model now mentioning the most relevant object rather than some secondary one when it had to choose (matching GPT-4o's choice).
- **Fluency:** It was high baseline and got even a bit higher (9.59). Likely some overly verbose or slightly awkward phrasing was eliminated. Essentially, the fine-tuned model writes very clean, simple sentences now -which are easy for GPT-4o (and humans) to read.



**Figure 7.** Comparison of the baseline and fine-tuned model on each evaluation metric

The fine-tuned model (orange) shows consistent improvements over the baseline (yellow) across all criteria. Notably, the largest gains are in “Hallucination” and “Instruction Following”, reflecting the model’s improved factual accuracy and adherence to the desired concise style. Minor improvements in other areas cumulatively contribute to a significantly higher final score.

In addition to numeric scores, we looked at some qualitative examples of the fine-tuned model’s output:



**Figure 8.** *Good SFT model answer 1*

For instance, an image with reference “Person wearing a colorful hat and glasses” was captioned by the fine-tuned model as “A person wearing a colorful hat and glasses”. This is nearly perfect: it matches the reference except for using “person” instead of “man” (which might even be preferred for neutrality). The caption is concise and covers the main subject and context. The judge likely gave this one very high scores in all categories (no hallucination, full coverage, semantically identical).



**Figure 9.** *Good SFT model answer 2.*

Another Good Example: The SFT model generated completely the same caption “Person cooking in a busy food stall.”

In summary, fine-tuning with LoRA on the GPT-4o dataset has clearly improved the image captioning performance according to our metrics and qualitative review. The model is more accurate, concise, and consistent with the desired alt-text style.

## 11. Inference Notebook for Easy Testing

To make it easy for the instructor to test the fine-tuned model, we prepared a Jupyter inference notebook (*src/run\_inference/Inference.ipynb*). This notebook is designed for convenient use on Google Colab or similar setups.

### Key points about the notebook:

- File: *src/run\_inference/Inference.ipynb* (with an easy link to open in Colab).
- Purpose: Allows the user to load the fine-tuned model (base Qwen3-VL-2B + our LoRA) and generate captions on any images, including a new dataset or custom uploads. It provides a simple interface: just pick or upload an image and run the captioning cell.
- Colab-Ready: The notebook includes a snippet to install necessary packages and to automatically download our fine-tuned LoRA weights from Hugging Face Hub. It also has a toggle or mention to ensure the runtime has a GPU (which is required for reasonable performance).
- Usage: By default, the notebook demonstrates on a new dataset *bodhisattamaiti/StyleExNet5k* (taking a small sample of 10% from it). This dataset contains stylish images (the exact content is not too important; we just wanted to show the model on a different domain). The notebook will randomly select an image from this dataset, display it, and then output the model's caption below it.
- Instructions: We included comments in the notebook for the instructor:
  - How to change the image or dataset (e.g., use `files.upload()` to upload a custom image).
  - Reminder to choose GPU hardware in Colab (Runtime > Change runtime type > Hardware Accelerator: GPU).
  - The expected behavior: the model should produce a caption within a few seconds per image. Since it's only 2B, it runs fairly quickly on modern GPUs.
- Example output: For instance, one test run in the notebook on a StyleExNet image (say a painting of a landscape) gave a caption "A painting of a landscape with mountains in the background." - showing the model can generalize its alt-text ability beyond photographs.

This notebook ensures that the evaluator of the project can quickly verify the model's performance on novel images without dealing with environment setup. It's essentially plug-and-play.

## 12. Conclusions

Through this project, we successfully improved an image captioning model’s performance by fine-tuning it on a tailored dataset. The fine-tuned model generates captions that are more concise, accurate, and aligned with the desired style (GPT-4o alt-text) than the initial baseline. Our evaluation using a GPT-4 judge model quantifies these improvements, and we observed a substantial increase in overall score and a large reduction in hallucinated details.

To highlight the progress, here is a comparison of the baseline vs fine-tuned scores:

| Metric                     | Baseline      | Fine-Tuned    | $\Delta$ Improvement |
|----------------------------|---------------|---------------|----------------------|
| Semantic Similarity        | 7.001         | 7.877         | +0.876               |
| Object Coverage            | 8.553         | 8.740         | +0.187               |
| Hallucination Score        | 7.869         | 9.191         | +1.322               |
| Completeness               | 7.627         | 8.022         | +0.395               |
| Instruction Following      | 7.422         | 8.966         | +1.544               |
| Fluency                    | 9.231         | 9.591         | +0.360               |
| <b>Final Score (0-100)</b> | <b>78.625</b> | <b>85.984</b> | <b>+7.359</b>        |

(Higher is better for all metrics. Improvement is Fine-Tuned minus Baseline.)

Every metric improved, with the largest gains in Hallucination and Instruction Following, which were our primary targets. The final score improvement of ~7.4 points (out of 100) is significant, demonstrating the efficacy of even a brief fine-tuning on a well-chosen dataset.

### Error Analysis

Despite the improvements, some limitations and error patterns remain:

- **Complex Scenes:** In images with many objects or people, the model’s single-sentence caption might not capture every detail. It prioritizes like the reference did, which sometimes means leaving out what a human might consider a secondary detail. For example, in a busy street scene, it might say “A bus on a street” focusing on the bus, missing that there are people around, if the reference also ignored them. This is inherent to the task: alt-texts are brief by design.
- **Ambiguities:** The model can still misinterpret actions or relationships if they are not obvious. If a person is holding a phone to their ear, is that “talking on the phone” or “holding a phone”? The model might choose one phrase and if the reference expected the other, it’s marked down. This isn’t a huge issue but shows the limit of pure supervised training – it can’t reason beyond the patterns it saw. Some ambiguity could be resolved by a more advanced multimodal reasoning model or by ensembling.
- **Small Objects and Background Elements:** If an important element is small or partially visible, the model sometimes omits it, possibly thinking it’s unimportant. E.g., a bird in the sky of an image might not be mentioned. The GPT-4o references often didn’t mention such things either, but if they did and our model didn’t, that’s a miss. This could be due to the model’s visual encoder limits (ViT might not

focus on tiny regions well) or the training giving more weight to large, central objects.

- **Stylistic Rigidity:** By aligning strongly to GPT-4o’s style, the model might be a bit too terse or neutral in cases where a bit more detail would be harmless. For instance, GPT-4o captions never say “wearing sunglasses” even if clearly visible, to keep it short. Our model also learned to often ignore such details. While this was desired for this project, in a broader captioning sense it’s a limitation (could be addressed by training on multiple styles or allowing a setting for detail level).

In conclusion, this project demonstrated how a modern VL model can be adapted with minimal training to meet specific task requirements. By leveraging a high-quality synthetic dataset and a powerful evaluation model, we achieved a strong alignment between the model’s outputs and the desired ground truth. The model now produces captions that are suitable as alt-text: short, accurate, and helpful.

[We provided all the code](#), results, and an easy-to-use inference notebook for verification. The improvements from baseline to fine-tuned model are clear, both in numbers and examples. With more time and resources, we could further refine the model, but even this result is a successful step towards reliable image captioning with limited computing resources.